

Project: Acoustic Gestures for Privacy-Preserving and Hands-Free Interaction in AI Chatbot Live Dialogue Modes

Problem Statement

Current Large Language Model voice interfaces, such as Gemini Live, rely heavily on two primary triggers: explicit touch and valid vocal inputs. These methods present limitations in the following scenarios:

- **Physical Constraints:** In "hands-busy" or "eyes-busy" situations, such as cooking, driving, or when the device is placed out of reach, manual touch is impossible or unsafe.
- **Privacy Concerns:** The "Always-listening" nature of voice assistants creates psychological discomfort, as users feel uneasy about a model continuously monitoring private conversations to detect valid voice input.

At this point, I propose a method to manage the settings and the actions of the voice assistants with non-private, reproducible sound patterns, such as knocking or clapping twice. (The patterns will be defined later.)



Fig 1. Gemini Live

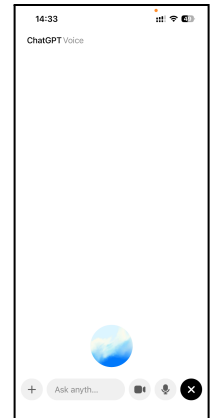


Fig 2. ChatGPT Voice

Related Works

- Y. Chen et al., "Gesture and Audio-Haptic Guidance Techniques to Direct Conversations with Intelligent Voice Interfaces," CHI 2025 - Proposes multimodal guidance techniques using gestures and audio-haptic feedback to allow users to efficiently manage dialogue flow in voice-based AI interactions.
- F. Bevilacqua et al., "A Meta-Model for Sonic Gestures in Collaborative Musical Performance," SMC 2011 - Establishes a structural model for "sonic gestures," providing a framework to understand how non-verbal acoustic patterns can convey intentional meaning and trigger system responses.
- E. Zeng et al., "Listen Only When Spoken To: Interpersonal Communication Cues as Smart Speaker Privacy Controls," PoPETs 2020 - Investigates the use of interpersonal communication cues as privacy-preserving triggers, demonstrating that non-verbal signals can effectively manage device "listening" states while reducing psychological discomfort from continuous monitoring.
- While these studies explore new triggers and privacy, they are not yet fully applied to LLM voice interfaces. Current LLMs still mostly rely on "always-listening" voice commands or physical touch, which are prone to privacy leaks, or impossible to use in hands-busy situations.

Proposed Method

This project proposes an interaction framework that utilizes Non-verbal Acoustic Gestures (e.g., claps, knocks, or whistles) as discrete, hands-free triggers for LLM control in Live Dialogue modes. By shifting the control mechanism from linguistic commands to acoustic patterns, we provide a less-constraint and privacy-conscious user experience with LLM voice interfaces.

Expected Output

The expected output for this project is a functional prototype: A mobile-based interface, a) possibly working as an assistant while using LLM voice interfaces or b) integrated with the Gemini API that responds in real-time to non-verbal acoustic triggers without physical touch. With the prototype, users would be able to do the following jobs with reproducible sound patterns:

- Resume/Stop Active Listening
- Adjust Speech Speed (Slower/Faster)
- Repeat the previous response
- Change the model